

Total No. of Questions : 4+4]

[Total No. of Printed Pages : 7

Roll No. _____

BS/Sem IV/107 (AB)

B. Sc. (Hons) Semester IV Examination

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CHEMISTRY

Paper No. CHB-401 : Inorganic Chemistry–II and Organic Chemistry–II

Time : 3 hours

Full Marks : 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks.

USE SEPARATE ANSWER BOOK FOR EACH SECTION

Section—A
(Inorganic Chemistry–II)
(Marks : 35)

Answer *three* questions including Question 1 which is compulsory

1. 1. What happens when CH_3COOH is dissolved in liquid NH_3 and H_2SO_4 ? Explain. **3**
 2. Why is alkali-metal liquid NH_3 solution deep blue and good reducing agent? **2**
 3. Why are Ti^{3+} complexes coloured while Ti^{4+} complexes are colourless? **2**
 4. Why is decrease in size from Ce to Lu very small? Explain. **2**
 5. Which one is a better Lewis acid BCl_3 or $\text{B}(\text{CH}_3)_3$? **2**
2. 1. Identify the Brønsted-Lowry acid, base, conjugate acid and conjugate base in the following : **3**
 - (i) $\text{NH}_4^+ + \text{S}^{2-} \rightleftharpoons \text{HS}^- + \text{NH}_3$
 - (ii) $\text{CO}_3^{2-} + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{OH}^-$
2. What are the characteristic species of acids and bases in BrF_3 and liquid NH_3 ? **2**
3. Giving three equations, illustrate Lewis acids and bases. **3**
4. Complete the reactions and identify the Lux-Flood acids and bases in the following : **2**
 - (i) $\text{Na}_2\text{O} + \text{SiO}_2 =$
 - (ii) $3\text{Na}_2\text{O} + \text{P}_2\text{O}_5 =$
5. In terms of HSAB principle which end of NCO^- would you expect to coordinate to Co^{2+} and Ti^{4+} ? **2**

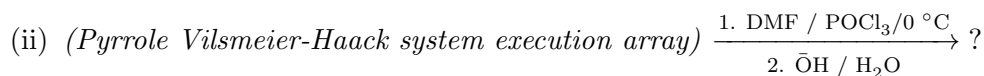
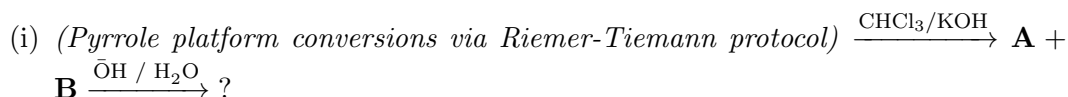
- Q.3.**
1. Why do transition metals exhibit variable oxidation states? How many oxidation states are exhibited by Fe and Cr and which one is more common? **3**
 2. Write IUPAC names of the following : **3**
 - (i) $[\text{Co}(\text{ONO})(\text{NH}_3)_5]^{2+}$
 - (ii) $[(\text{NH}_3)_5\text{Cr}-\text{OH}-\text{Cr}(\text{NH}_3)_5]\text{Cl}_5$
 - (iii) $\text{K}_2[\text{Zn}(\text{NH}_2)_4]$
 3. Detect the structure and mention the name of isomers of $[\text{Co}(\text{en})_2\text{Cl}_2]^+$. **3**
 4. Calculate the effective atomic number of the metal in the following : **3**
 - (i) $\text{Fe}(\text{CO})_5$
 - (ii) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 - (iii) $\text{Co}_2(\text{CO})_8$
- 4.**
1. Name some properties that mainly determine the utility of a solvent. **3**
 2. Giving two examples compare and contrast the precipitation reactions in aqueous and liquid NH_3 . **3**
 3. Explain the important consequences of lanthanide contraction. **3**
 4. Write outer electronic configuration of Ho and Lu in the ground state and indicate number of unpaired electron in Ho^{3+} and Lu^{3+} ions. **3**
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Section—B
(Organic Chemistry—II)
(Marks : 35)

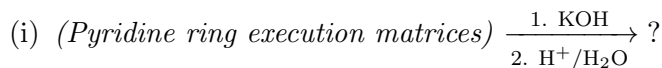
*Answer **three** questions including Question 1 which is compulsory*

- 1.**
1. Pyridine is more basic than pyrrole but less basic than aliphatic tertiary amines. Explain. **2**
 2. Write the cyclic structures of D (+) glucose. **2**
 3. How will you prepare indigotin from anthranilic acid? **2**
 4. Complete the following reactions : **3×1+2**
 - (i) (*Anthracene + Maleic Anhydride system*) $\xrightarrow{\Delta}$?
 - (ii) (*Pyridine layout substitution array*) $\xrightarrow[\Delta]{\text{NaNH}_2}$?
 - (iii) (*Furan gas-phase layout conversions*) $\xrightarrow{\text{HCN} / \text{HCl} / \text{H}_2\text{O}}$?
 - (iv) (*Phthalic anhydride + Benzene architecture*) $\xrightarrow{\text{AlCl}_3}$? $\xrightarrow[2. \text{Zn}]{1. \text{c.H}_2\text{SO}_4}$?
- 2.**
1. Give a reaction sequence to convert glucose into fructose. **4**

2. Write the name of the following reactions and explain the formation of products with suitable mechanism : **3×2**



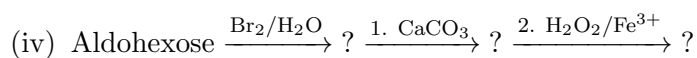
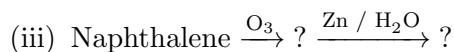
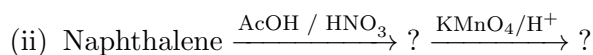
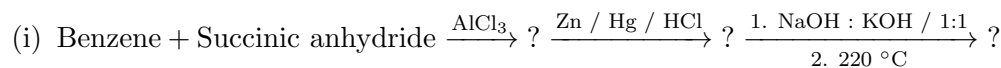
3. Complete the following reaction : **2**



3. 1. Describe Hantzsch synthesis for pyridine. **6**

2. Describe the structure of indigotin. **6**

4. Predict the products of following reactions : **4+(2×2)+4**



External Portals & Academic Resources:

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app>